

$$\lim_{x \rightarrow 0} \frac{e^{3x} - 1 - 3x}{x^2} = \lim_{x \rightarrow 0} \frac{(e^{3x} - 1 - 3x)'}{(x^2)'} =$$

$$\left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{3e^{3x} - 3}{2x} = \frac{3}{2} \lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x} =$$

$$= \frac{3}{2} \lim_{x \rightarrow 0} \frac{(e^{3x} - 1)'}{(x)'} = \frac{3}{2} \lim_{x \rightarrow 0} \frac{3e^{3x}}{1} =$$

$$= \frac{9}{2} \cdot \lim_{x \rightarrow 0} e^{3x} \rightarrow \left(\frac{9}{2}\right)$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^{3x} = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\frac{x}{2}}\right)^{3x} =$$

$$= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{1}{\frac{x}{2}}\right)^{\frac{x}{2}} \right)^{\frac{2}{x} \cdot 3x} \rightarrow e^6$$



$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \rightarrow e$$

$$\int_{-2}^3 x \sqrt{7-x} dx \quad (\equiv)$$

$$t = 7 - x, x = 7 - t$$

$$dt = d(7 - x) = (7 - x)' dx = -dx$$

$$dx = -dt$$

$$\int -(7 - t) \sqrt{t} dt = \int (t - 7) \sqrt{t} dt =$$

$$= \int t \sqrt{t} dt - 7 \int \sqrt{t} dt = \int t^1 \cdot t^{\frac{1}{2}} dt - 7 \int t^{\frac{1}{2}} dt =$$

$$= \frac{t^{\frac{3}{2}+1}}{\frac{3}{2}+1} - \frac{7 t^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{2 t^{\frac{5}{2}}}{5} - \frac{14 t^{\frac{3}{2}}}{3} = \frac{6 t^{\frac{5}{2}} - 70 t^{\frac{3}{2}}}{15} =$$

$$= \frac{6(7-x)^2 \sqrt{7-x} - 70(7-x) \sqrt{7-x}}{15} =$$

$$= \frac{2(7-x) \sqrt{7-x} (3(7-x) - 35)}{15} =$$

$$= \frac{2(7-x) \sqrt{7-x} (-3x - 14)}{15} + C$$

$$(\equiv) \left. \frac{2(7-x) \sqrt{7-x} (-3x - 14)}{15} \right|_{-2}^3 =$$

$$= \frac{2(7-3) \sqrt{7-3} (-3 \cdot 3 - 14) - 2(7-(-2)) \sqrt{7-(-2)} (-3(-2) - 14)}{15}$$

$$= \frac{2 \cdot 4 \cdot 2 (-23) - 2 \cdot 9 \cdot 3 (-8)}{15} = \frac{-368 + 432}{15} =$$

$$= \frac{64}{15}$$

$$\begin{vmatrix} 1 & 2 & -1 & 2 \\ 0 & 2 & 5 & 3 \\ 3 & 2 & -1 & 4 \\ 4 & 3 & 0 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1 & 2 \\ 0 & 2 & 5 & 3 \\ 3 & 0 & -1 & 4 \\ 4 & 3 & 0 & 2 \end{vmatrix} = 12 \begin{vmatrix} 1 & -1 & 2 \\ 3 & -1 & 4 \\ 4 & 0 & 2 \end{vmatrix} + 3 \begin{vmatrix} 1 & -1 & 2 \\ 0 & 5 & 3 \\ 3 & -1 & 4 \end{vmatrix} =$$

$$= 12(4 \begin{vmatrix} -1 & 2 \\ -1 & 4 \end{vmatrix} + 2 \begin{vmatrix} 1 & -1 \\ 3 & -1 \end{vmatrix}) + 3(\begin{vmatrix} 5 & 3 \\ -1 & 4 \end{vmatrix} + 3 \begin{vmatrix} -1 & 2 \\ 5 & 3 \end{vmatrix}) =$$

$$= 12(4(-4+2) + 2(-1+3)) + 3((20+3) + 3(-3-10)) =$$

$$= 12(-8+4) + 3(23-39) = -48-48 = -96$$